

Mapping Lifecycle Metadata Ontology & Mapping Reuse

This self-administered survey is designed to evaluate a lifecycle-based metadata ontology (MMV) and to assess whether lifecycle metadata improves experts' decisions about reusing declarative mappings. The survey also includes a second section exploring whether modelling "purpose" as a controlled vocabulary could further enhance interpretation, governance, and reuse of mapping artefacts.

1. I consent to participate in this study.

Purpose. The study evaluates a lifecycle-based metadata ontology and tests whether lifecycle metadata supports mapping reuse decisions. Participation is voluntary and responses are anonymous. The questionnaire takes approximately **15-30 min** to complete.

Note: All responses are anonymized and used only for academic research. You may stop at any time by closing the form. By proceeding you confirm you are at least 18 years old and consent to participate.

Mark only one oval.

☐ Yes

☐ No

Introduction and Illustrative Example

The Mapping Metadata Vocabulary (MMV) is a lifecycle-based ontology designed to document declarative mapping projects (e.g., RML mappings, ontology alignments, or interlinking rules).

MMV structures metadata across five lifecycle phases:

- Analysis : purpose, requirements, input description, assumptions
- Design : modelling decisions and justifications
- Development : implementation details
- Testing : validation results and quality assessment
- Maintenance : versioning and publishing information

The goal of MMV is to improve mapping transparency, documentation quality, evaluation, and reuse.

The example below illustrates how MMV can be used to document a mapping project. You may refer to this example while reviewing the ontology.

The model specification and ontology are accessible via this [Link](#)

Documentation of ontology scope, lifecycle phases, competency questions and ontology design can be found in our latest published [paper](#)

Illustrative MMV example (Turtle representation).

The example below shows a mapping project documented using MMV. It demonstrates how a MappingProject links to lifecycle phases (Analysis, Design, Testing, Maintenance). Each phase records specific metadata (e.g., purpose, design decisions, testing results). Stakeholders and quality metrics are explicitly represented. In this example:

- The project (ex:Project1) is connected to its lifecycle phases.
- The Analysis phase records the mapping purpose and input dataset.
- The Design phase records modelling decisions and planned quality metrics.
- The Testing phase captures validation results.
- The Maintenance phase records publishing information.

You do not need to analyse the syntax in detail. The example is provided to illustrate how MMV structures mapping metadata.

```

@prefix mmv: <http://w3id.org/mmv/0.1#> .
@prefix prov: <http://www.w3.org/ns/prov#> .
@prefix dqv: <http://www.w3.org/ns/dqv#> .
@prefix ex: <http://example.org/> .

ex:Project1 a mmv:MappingProject ;
  mmv:hasAnalysis ex:Analysis1 ;
  mmv:hasDesign ex:Design1 ;
  mmv:hasTesting ex:Test1 ;
  mmv:hasMaintenance ex:Maintenancel ;
  mmv:mmvhasStakeholder ex:Sarah .

ex:Analysis1 a mmv:AnalysisActivity ;
  mmv:hasPurpose "Transform CSV dataset into RDF for publication" ;
  mmv:hasMappingDomain "Geospatial" ;
  mmv:hasRequirement "Comply with GeoSPARQL" ;
  mmv:hasInputName "ElectoralDivisions2022.csv" ;
  mmv:hasAssumption "Coordinates are provided in WGS84" .

ex:Design1 a mmv:DesignActivity ;
  mmv:hasDesignDecision "Reuse GeoSPARQL vocabulary for geometry representation" ;
  mmv:hasJustification "Ensures interoperability with geospatial Linked Data tools" ;
  mmv:hasPlannedQualityMetric ex:CompletenessMetric .

ex:CompletenessMetric a dqv:Metric .

ex:Test1 a mmv:TestingActivity ;
  mmv:hasTestingType "SHACL validation" ;
  mmv:hasTestingResult "All constraints passed" .

ex:Maintenancel a mmv:MaintenanceActivity ;
  mmv:hasPublisherName "Trinity College Dublin" .

ex:Sarah a prov:Agent ;
  mmv:hasRole "Mapping Developer" .

```

Based on the description, example, and documentation provided, please indicate your level of agreement with the following statements.

2. 1- The ontology captures the essential concepts and relationships required to document mapping projects across the lifecycle.

Mark only one oval.

	1	2	3	4	5	
Stro	☐	☐	☐	☐	☐	Strongly Disagree

3. 2- The overall structure of the ontology is coherent and easy to understand.

Mark only one oval.

	1	2	3	4	5	
Stro	☐	☐	☐	☐	☐	Strongly Disagree

4. 3- The ontology classes and properties are clearly defined and non-ambiguous.

Mark only one oval.

	1	2	3	4	5	
Stro	☐	☐	☐	☐	☐	Strongly Disagree

5. 4- The ontology follows recognised ontology design principles (e.g., clarity, coherence, reuse).

Mark only one oval.

	1	2	3	4	5	
Stro	☐	☐	☐	☐	☐	Strongly Disagree

6. 5- The ontology appropriately reuses existing and well-established vocabularies.

Mark only one oval.

	1	2	3	4	5	
Stro	☐	☐	☐	☐	☐	Strongly Disagree

7. 6- The ontology would support informed and confident decisions about mapping reuse.

Mark only one oval.

	1	2	3	4	5	
Stro	☐	☐	☐	☐	☐	Strongly Disagree

8. 7- Do you suggest any modifications to the concepts, properties or modelling decisions ? Please justify.

9. 8- Do you suggest improvements to the documentation (examples, explanations, constraints)? Please justify your response.

Purpose Vocabulary

We are exploring whether “purpose” should be modelled as a first-class metadata element supported by a controlled vocabulary to improve reuse, interpretation, and governance of mapping artefacts. The following questions explore whether modelling “purpose” explicitly could strengthen mapping interpretation and reuse.

10. 1- Do you think the purpose of a mapping artefact should be explicitly captured using a controlled vocabulary?

Mark only one oval.

- ☐ Yes, strongly agree
- ☐ Yes, agree
- ☐ Neutral / unsure
- ☐ Disagree
- ☐ Strongly disagree

11. 2- Which of the following purpose categories do you consider useful for describing mapping artefacts (*Select all that apply*)

Check all that apply.

- ☐ 1- Project Goal (e.g., data integration, enrichment)
- ☐ 2- Application Context or Domain (e.g., research, operational system, decision support)
- ☐ 3- Intended Use (e.g., analysis, publication, data exchange)
- ☐ 4- Target Audience (e.g., domain experts, developers, public users)
- ☐ 5- Reuse Expectations (e.g., exploratory reuse, production reuse, one-off use)

12. 3- Rank the importance of the following purpose categories (1 = Most Important, 5 = Least Important)

Mark only one oval per row.

	1	2	3	4	5
Project Goal	☐	☐	☐	☐	☐
Application Context or Domain	☐	☐	☐	☐	☐
Intended Use	☐	☐	☐	☐	☐
Target Audience	☐	☐	☐	☐	☐
Reuse Expectations	☐	☐	☐	☐	☐

13. 4- Do you have suggestions for additional purpose categories, or comments on how purpose metadata could support reuse and interpretation of mappings?

14. 5- Do you think a controlled vocabulary for describing the purpose of an artefact should be applicable beyond mapping projects, to Semantic Web projects in general (e.g., datasets, ontologies, workflows)?

Mark only one oval.

- ☐ Yes, strongly agree
- ☐ Yes, agree
- ☐ Neutral / unsure
- ☐ Disagree
- ☐ Strongly disagree

15. 6- if yes, please describe how a general purpose-oriented vocabulary could be used in practice (e.g., to support reuse, interpretation, governance, or decision-making).

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